# Doeun Kim

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## RESEARCH INTERESTS

Machine Learning, AI for Healthcare and Biomedicine, Explainable AI, Multimodal Learning, Generative Models

## EDUCATION

**University of Southern California** Los Angeles, CA
M.S. in Computer Science June 2025 – Present (Expected December 2026)

- GPA: 3.82 / 4.0
- Relevant coursework: Machine Learning, Deep Learning, Applied NLP, Information Retrieval & Web Search, Algorithms

**Ewha Womans University** Seoul, South Korea
B.S. in Computer Science and Engineering & B.S. in Content Convergence February 2025

- GPA: 4.21 / 4.3 (Top 4%, *Summa Cum Laude*; Dean's List, all semesters)
- Undergraduate thesis: *Layer-Adaptive Quantization of Diffusion Models using Fisher Information*
- Relevant coursework: Data Structures, Algorithms, Computer Architecture, Operating Systems, Database, Software Engineering

**San José State University** San José, CA
International Student Exchange Program, majoring in Computer Science August 2022 – December 2022

- Relevant coursework: Introduction to AI, Formal Language and Computability

## RESEARCH EXPERIENCE

**Laboratory for Machine Learning, Health and Biomedicine @ University of Southern California** Los Angeles, CA
Graduate Research Assistant (Advisor: Prof. Ruishan Liu) November 2025 – Present

- Developed a deimmunization algorithm for synthetic binders, identifying amino acid substitutions in high-immunogenicity peptide regions aimed at reducing predicted immunogenicity while preserving binding affinity
- Optimized deimmunization search with peptide-level score caching and beam search, reducing redundant immunogenicity evaluations while retaining multiple promising candidate substitutions for sequence-level optimization
- Built a FastAPI- and Gradio-based research interface for HLA-specific peptide scanning, allowing users to configure target peptides, HLA alleles, and presentation thresholds and inspect matched peptide outputs in tabular form
- Implemented asynchronous job orchestration with RabbitMQ and singleton-managed model services to support concurrent peptide scans and immunogenicity scoring by HLA allele and population group

**AI Computing Platform Lab @ Ewha Womans University** Seoul, South Korea
Undergraduate Researcher (Prof. Jaehyeong Sim) April 2024 – February 2025

- Led an undergraduate research team on Layer-Adaptive Quantization of Diffusion Models, using Fisher Information to measure layer-wise contribution and guide stronger quantization of less critical weights while maintaining image quality
- Reduced model size by 49.4% and improved FID by 7% through Fisher-guided layer-wise quantization; presented the work at the 2024 IEIE Symposium as a co-first author and presenter
- Led a DDPM interpretability study using Integrated Gradients, GradSHAP, and Occlusion on ImageNet-1000 to examine which visual regions the model emphasized at different generation timesteps
- Found that DDPM generation progressed from global outlines to salient features and fine details, indicating a human-like sequential drawing process; presented the work at the KCC 2024 XAI Workshop as the only undergraduate-led team

## PUBLICATIONS & PRESENTATIONS

- **Kim, D**., Byeon, J., Park, I., & Sim, J. *Layer-Adaptive Quantization on Diffusion Models using Fisher Information*. Poster presented at the 2024 Symposium of the Institute of Electronics and Information Engineers, Seoul, South Korea, December 27, 2024. Co-first author and presenter.
- **Kim, D**., Byeon, J., & Park, I. *Analysis of Diffusion Model Inference Mechanisms Using XAI Techniques*. Poster presented at the KCC 2024 XAI Workshop, Jeju, South Korea, June 27, 2024. Co-first author and presenter.

## ACADEMIC PROJECTS

**Multimodal Entrance Detection System** January 2026 – Present

- Contributed to DoRA-based Qwen fine-tuning for multimodal entrance detection using geospatial inputs, including aerial imagery, street-view images, GPS trajectories, and building footprint overlays
- Conducted API-based in-context learning experiments and comparative VLM prediction analysis, and developed GPS-aware reranking across 64 candidate entrance locations, improving F1 from 0.30 to 0.59 and AUPR from 0.17 to 0.45

**PokerBench – SLMs for Poker Decision-Making** August 2025 – December 2025

- Fine-tuned small language models, including Qwen3, LLaMA-3.2, and Gemma variants, using QLoRA and LoRA adapters on approximately 560K solver-labeled poker decision-making samples
- Built an evaluation pipeline for instruction-following under uncertainty and compared model performance against larger proprietary baselines using Action Accuracy and Exact Match

**TalesRunner – AI-powered Storybook Generation from Images** January 2025 – February 2025

- Developed a multimodal generation pipeline combining BLIP-based image captioning, fine-tuned KoT5 story generation, and GPT-4o-assisted text refinement to generate personalized storybooks from user-provided images
- Integrated generated captions, story text, and user images into downloadable PDF storybooks, validating narrative quality, image-text alignment, and formatting consistency across 100+ test cases

**Jeans – AI-Based Photo Editing and Sharing Service for seniors** December 2024 – February 2025

- Led a 5-member team in developing AI features for a senior-focused photo editing and sharing service, including fine-tuned Whisper for Korean dialect speech recognition and YOLO/FaceNet-based facial recognition
- Deployed FastAPI-based ML inference services on AWS EC2 with GitHub Actions CI/CD, refining speech-driven interaction flows and accessibility through senior user interviews and usability tests

## PROFESSIONAL EXPERIENCE

**RainbirdGEO** Seoul, South Korea
Software Engineer Intern, Full-stack May 2021 – July 2022

- Built full-stack features for a weather-forecasting mobile application using React Native and Node.js/Express, supporting pilot deployment and usability testing for users in Southeast Asian regions
- Translated requirements analysis, competitor research, and user feedback into functional specifications, interface documentation, and iterative product updates
- Analyzed 2,000+ survey responses from 100+ users with Tableau dashboards, contributing to improvements in user satisfaction from 2.9 to 4.1 and forecast accuracy from 63% to 96%

## TEACHING & MENTORING

**Women in Engineering Mentor** Los Angeles, CA
USC Viterbi School of Engineering January 2026 – Present

- Support women graduate students through peer mentoring, academic guidance, and discussions on navigating engineering graduate programs

**Viterbi Graduate Mentor** Los Angeles, CA
USC Viterbi School of Engineering September 2025 – Present

- Mentor incoming graduate students from diverse academic and cultural backgrounds on course planning, research involvement, and adjustment to graduate student life at USC

**Instructional Assistant (Part-time)** Seoul, South Korea
Scholar English – Private English Academy September 2023 – May 2025

- Supported large-group English instruction at a private academy through grading, assignment review, instructional material preparation, curriculum revision, and student communication
- Developed and maintained a custom GPT-based chatbot to streamline instructional workflows and support student-facing communication

**Volunteer Tutor** Seoul, South Korea
Neulhaerang School – Inter-University Volunteering Org. (Ewha W. Univ. & SNU) March 2021 – December 2021

- Designed accessible learning materials and taught weekly computer literacy classes for elementary students through an inter-university volunteer education program

## HONORS & AWARDS

National Scholarship for Science and Engineering, KOSAF (Full tuition scholarship, 2 years) 2022 – 2024
Capstone Project Merit Award, Ewha Womans University 2024
Individual Best Award, SK Telecom FLY AI Program (run by South Korea's largest telecom company) 2025
Outstanding Project Award, SK Telecom FLY AI Program 2025
Honors Scholarship, Ewha Womans University (Awarded 4 times) 2020 – 2024
International Study Scholarship, Ewha Womans University 2022

## TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Scikit-learn, Keras, OpenCV, Hugging Face Transformers, LoRA/QLoRA, BLIP, Whisper
Research & Data: NumPy, Pandas, Tableau, model evaluation, prompt formulation, dataset preprocessing, experimental analysis
Languages: Python, Java, C, C#, SQL, JavaScript/TypeScript, HTML/CSS
Backend & ML Systems: FastAPI, Django, Express, Spring, React, React Native, Gradio, REST APIs
Cloud & DevOps: AWS, MS Azure, Docker, Git, GitHub Actions, CI/CD
Databases: MySQL, MongoDB, MariaDB, SQLite